Holistic Evaluation of Language Models

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Abstract

Language models (LMs) are becoming the foundation for almost all major language technologies, but their capabilities, limitations, and risks are not well understood. We present Holistic Evaluation of Language Models (HELM) to improve the transparency of language models. First, we taxonomize the vast space of potential scenarios (i.e. use cases) and metrics (i.e. desiderata) that are of interest for LMs. Then we select a broad subset based on coverage and feasibility, noting what's missing or underrepresented (e.g. question answering for neglected English dialects, metrics for trustworthiness). Second, we adopt a multi-metric approach: We measure 7 metrics (accuracy, calibration, robustness, fairness, bias, toxicity, and efficiency) for *each* of 16 core scenarios to the extent possible (87.5% of the time), ensuring that metrics beyond accuracy don't fall to the wayside, and that trade-offs across models and metrics are clearly exposed. We also perform 7 targeted evaluations, based on 26 targeted scenarios, to more deeply analyze specific aspects (e.g. knowledge, reasoning, memorization/copyright, disinformation). Third, we conduct a large-scale evaluation of 30 prominent language models (spanning open, limited-access, and closed models) on all 42 scenarios, including 21 scenarios that were not previously used in mainstream LM evaluation. Prior to HELM, models on average were evaluated on just 17.9% of the core HELM scenarios, with some prominent models not sharing a single scenario in common. We improve this to 96.0%: now all 30 models have been densely benchmarked on a set of core scenarios and metrics under standardized conditions. Our evaluation surfaces 25 top-level findings concerning the interplay between different scenarios, metrics, and models. For full transparency, we release all raw model prompts and completions publicly[1] for further analysis, as well as a general modular toolkit for easily adding new scenarios, models, metrics, and prompting strategies.[2] We intend for HELM to be a living benchmark for the community, continuously updated with new scenarios, metrics, and models.

† indicates lead authors and ‡ indicates major contributors. Full author contributions in Appendix A.
[1] https://crfm.stanford.edu/helm/v0.1.0
[2] https://github.com/stanford-crfm/helm

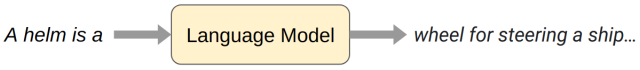

Figure 1: **Language model.** A language model takes text (a prompt) and generates text (a completion) probabilistically. Despite their simple interface, language models can be adapted to a wide range of language tasks from question answering to summarization.

1 Introduction

Benchmarks orient AI. They encode values and priorities (Ethayarajh & Jurafsky, 2020; Birhane et al., 2022) that specify directions for the AI community to improve upon (Spärck Jones & Galliers, 1995; Spärck Jones, 2005; Kiela et al., 2021; Bowman & Dahl, 2021; Raji et al., 2021). When implemented and interpreted appropriately, they enable the broader community to better understand AI technology and influence its trajectory.

In recent years, the AI technology that has arguably advanced the most is foundation models (Bommasani et al., 2021), headlined by the rise of language models (LMs; Peters et al., 2018; Devlin et al., 2019; Brown et al., 2020; Rae et al., 2021; Chowdhery et al., 2022). At its core, a language model is a box that takes in text and generates text (Figure 1). Despite their simplicity, when these models are trained on broad data at immense scale, they can be adapted (e.g. prompted or fine-tuned) to myriad downstream scenarios. Yet the immense surface of model capabilities, limitations, and risks remains poorly understood. The rapid development, rising impact, and inadequate understanding demand that we benchmark language models holistically.

But what does it mean to benchmark language models holistically? Language models are general-purpose text interfaces that could be applied across a vast expanse of scenarios. And for each scenario, we may have a broad set of desiderata: models should be accurate, robust, fair, efficient, and so on. In fact, the relative importance of these desiderata often will depend not only on the perspective and values one has, but the scenario itself (e.g. inference efficiency might be of greater importance in mobile applications).

We believe holistic evaluation involves three elements:

1. **Broad coverage and recognition of incompleteness.** Given language models' vast surface of capabilities and risks, we need to evaluate language models over a broad range of scenarios. Broadening the evaluation has been a continuing trend in the NLP community, going from individual datasets such as SQuAD (Rajpurkar et al., 2016) to small collections of datasets such as SuperGLUE (Wang et al., 2019b) to large collections of datasets such as the GPT-3 evaluation suite (Brown et al., 2020), Eleuther AI LM Harness (Gao et al., 2021b), and BIG-Bench (Srivastava et al., 2022). However, it is neither possible to consider all the scenarios nor all the desiderata that (could) pertain to LMs. Therefore, holistic evaluation should provide a top-down taxonomy and make explicit all the major scenarios and metrics that are missing.

2. **Multi-metric measurement.** Societally beneficial systems reflect many values, not just accuracy. Holistic evaluation should represent these plural desiderata, evaluating every desideratum for each scenario considered.

3. **Standardization.** Our *object* of evaluation is the language model, not a scenario-specific system. Therefore, in order to meaningfully compare different LMs, the strategy for adapting an LM to a scenario should be controlled for. Furthermore, each LM should be evaluated on the same scenarios to the extent possible.

Overall, holistic evaluation builds transparency by assessing language models in their totality. Rather than honing in on a specific aspect, we strive for a fuller characterization of language models to improve scientific understanding and orient societal impact.

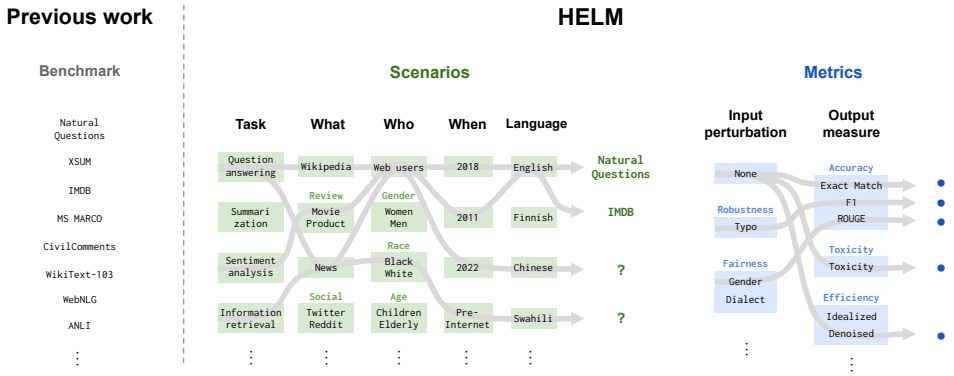

Figure 2: **The importance of the taxonomy to HELM.** Previous language model benchmarks (e.g. SuperGLUE, EleutherAI LM Evaluation Harness, BIG-Bench) are collections of datasets, each with a standard task framing and canonical metric, usually accuracy (*left*). In comparison, in HELM we take a top-down approach of first explicitly stating what we want to evaluate (i.e. scenarios and metrics) by working through their underlying structure. Given this stated taxonomy, we make deliberate decisions on what subset we implement and evaluate, which makes explicit what we miss (e.g. coverage of languages beyond English).

1.1 HELM

Holistic Evaluation of Language Models (HELM) has two levels: (i) an abstract taxonomy of *scenarios* and *metrics* to define the design space for language model evaluation and (ii) a concrete set of implemented scenarios and metrics that were selected to prioritize coverage (e.g. different English varieties), value (e.g. user-facing applications), and feasibility (e.g. limited engineering resources).

Recognition of incompleteness. Benchmarks across AI, including those for language models like SuperGLUE (Wang et al., 2019a), the EleutherAI LM Harness (Gao et al., 2021b), and BIG-bench (Srivastava et al., 2022), are defined by specific choices of scenarios and metrics. Different benchmarks make different decisions on what to prioritize, how to make these decisions, and to what extent these processes are made clear in presenting the benchmark. Since our aim is holistic evaluation, we believe it is necessary to be explicit on the relationship between what we aspire to evaluate and what we actually evaluate. The construction of HELM starts top-down with a taxonomy over scenarios and metrics (see Figure 2). The taxonomy not only facilitates the systematic selection of scenarios and metrics, but it also make explicit what is missing. We view HELM as a living benchmark, and we hope that both the abstract taxonomy and the concrete selection of scenarios and metrics will evolve according to the technology, applications, and social concerns. In §10: MISSING, we explicitly highlight evaluations HELM lacks that should be prioritized. Often these are ones the entire AI field has historically neglected.

Multi-metric measurement. HELM currently implements a core[3] set of 16 scenarios and 7 (categories of) metrics. Our scenarios, which are triples of (task, domain, language), span 6 user-facing tasks (e.g. question answering, information retrieval, summarization, toxicity detection), several domains (e.g. news, books), and currently only English (though we cover several English varieties such as African-American English and the English varieties spoken in different English-speaking countries). And our 7 categories of metrics reflect a range of societal considerations (i.e. accuracy, calibration, robustness, fairness, bias, toxicity, efficiency). We emphasize that while we have specific quantitative metrics for all of these considerations, they (e.g. fairness) are complex and contested social constructs that can be operationalized in many different ways. Consistent with our second element of holistic evaluation, we ensure our benchmark attains **dense** multi-metric measurement: of the 112 possible (core scenario, metric) pairs, we measure 98 (87.5%) as shown in Table 4.

[3]We use the term *core* to indicate that for this set of scenarios, we measure a range of metrics/desiderata. The term *core* is not meant to suggest that any specific scenario in this set is more fundamental than scenarios outside the set.